

VIBHAV INNA KEDEGE

MSc Embedded Systems (Honours), BSc Electrical Engineering

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📍 Rijswijk, The Netherlands

- Date of Birth: 27th July 1995
- Nationality: Dutch (Netherlands, EU) & Indian (OCI)



PROFESSIONAL EXPERIENCE

Alfen N.V.

DevOps Software Engineer | R&D Department

📅 Oct 2023-Present 📍 Almere, The Netherlands

- **Architected a Distributed Test Automation Platform** using Python, Jenkins, and Docker on Azure that accelerated firmware delivery cycles by **6x** (from 12 weeks to 2 weeks).
- **Pioneered a Real-time Analytics Engine** (FastAPI, PostgreSQL) to automate **2,000+ daily tests** analysis, that enables multiple teams identify firmware defects 10 weeks earlier, reclaiming **€150K per cycle** in engineering capacity.
- **Spearheading R&D for an MLOps Ecosystem** focusing on Predictive Maintenance and LLM-based Log Analysis to automate root-cause identification and de-risk technical validation for IoT battery systems.

KPN (via Capgemini)

Data Engineer (Big Data) | KPN In-Home Analytics

📅 Oct 2021-Sep 2023 📍 The Hague, The Netherlands

- **Engineered Scalable Data Pipelines** using Spark and Scala to ingest and monitor telemetry from **1M+ IoT devices** nationwide, processing billions of daily records for real-time network health analysis.
- **Developed a Predictive ML System** using Spark MLlib (Random Forest) to analyze real-time health metrics. This enabled proactive network management and significantly reduced customer churn.
- **Optimized Data Visualization Architectures** in Tableau and Python to provide executive-level insights into internet quality, transforming raw device logs into actionable business intelligence.

Almende B.V.

Robotics R&D Engineer | R&D Department

📅 Jun 2020-Aug 2021 📍 Rotterdam, The Netherlands

- **Developed Distributed Planning Algorithms** for multi-robot SLAM using Python, C++, and ROS that validated swarm coordination via high-fidelity simulations for autonomous fleet deployments. (Link)
- **Published Peer-Reviewed Research** on multi-robot path planning that translated complex algorithmic findings into a technical case study presented at (Link)

Alstom Transport

System Engineer - Train Networks | Train Control Department

📅 Jul 2017-Jul 2019 📍 Bangalore, India and Sydney, Australia

- **Integrated Ethernet-based Train Control Systems (TCMS)** for autonomous metros, achieving a **90% pass rate** and completing field validation in India and Australia **3 months ahead of schedule**.

PERSONAL STATEMENT

"Software Engineer with 7+ years of expertise in **Data Platforms** and **MLOps**. Specialized in accelerating delivery cycles by **6x** and architecting distributed systems for **1M+ devices** through high-performance engineering."

EDUCATION

Delft University of Technology

MSc Embedded Systems (Honours)

📅 2019 - 2021 📍 Delft, Netherlands

- Specialisation - Artificial Intelligence (AI) and Robotics.
- Thesis - "Multi Robot Exploration and Planning in limited communication environments" (Link).
- Published at ICAART 2022 (Link).

National Institute of Technology
Karnataka (NITK-Surathkal)

Bachelor of Technology, Electrical and Electronics Engineering

📅 2013 - 2017 📍 Mangalore, India

- Specialisation - Embedded Software
- Thesis - "Multiple Interface Automation using Distributed Embedded Devices" (Link)

TECHNICAL SKILLS

Core Languages

Python C++ SQL Scala

Data & AI

MLOps Apache Spark PyTorch

Vector Databases LLMs RAG

Platform & Cloud

Docker Jenkins Azure FastAPI

Linux

Engineering

Distributed Systems System Design

HIL Testing CI/CD Product Thinking

SELECTED PROJECTS

Predictive Stock Analytics Pipeline

Technical Case Study

📅 Jan 2025 – Present

- Engineered a **cloud-native analytics platform** to ingest S&P 500 market data using **Python and Docker**, help with market analysis and facilitate predictive modeling for trend forecasting. ([Live Demo](#))
- Optimized Data Visualization** via Streamlit to facilitate high-frequency correlation analysis and volatility tracking across major equities.

Local RAG & LLM Orchestration Engine

Technical Case Study

📅 Jul 2025

- Developed a **RAG pipeline** using **LangChain and Ollama** for local LLM inference; integrated a Vector Database for context-aware semantic retrieval. ([Technical Article](#))
- Architected context-injection logic** to minimize LLM hallucinations, validating a Retrieval-Augmented Generation (RAG) approach for secure, private data processing.

MLOps for Image Segmentation

Freelance Project

📅 Apr 2023 - Jul 2023

📍 Delft, Netherlands

- Architected an MLOps Pipeline** to automate training and deployment of **YOLOv8 and SAM** models using **Python**, reducing model iteration cycles from weeks to daily updates.
- Engineered Workflows for 100K+ high-resolution images**; optimized inference and deployment for logistics automation, bridging vision research with production-scale industrial needs.

RL Sensitivity & Optimization Analysis

Technical Case Study

📅 Sep 2020

📍 Delft, NL

- Developed and Benchmarked RL Agents** (SARSA, Q-Learning) within OpenAI Gym by performing multi-dimensional sensitivity analysis to identify optimal hyperparameter convergence.
- Evaluated Algorithmic Efficiency** across discrete environments and documented findings in a technical study. ([Technical Article](#))

HONORS & LEADERSHIP

- Alumni Advisor, ISA TU Delft**: Mentoring international students on navigating the Dutch higher education system and successfully transitioning into the tech industry.
- Visionary Engineer Award (Alstom Transport)**: Awarded for excellence in project planning and the optimization of system delivery schedules.
- President, ISTE (NITK Chapter)**: Led a 100+ member technical student society; managed large-scale engineering workshops and symposiums for 1,000+ participants.
- Community Volunteer**: Organized STEM-focused educational outreach for underprivileged students in Indian government schools to bridge the digital divide.

CERTIFICATIONS

- Advanced Python Development Techniques ([LINK](#))
- Google Advanced Data Analytics Specialization ([LINK](#))
- DeepLearning.AI TensorFlow Developer Specialization ([LINK](#))
- Professional Scrum Master I ([LINK](#))

MANAGERIAL SKILLS

Project Planning

Collaboration

Customer Interaction

Agile Working

Scrum

Mentoring and Training

KNOWN LANGUAGES

English

Dutch (A2)

German (A1)

Hindi

Kannada

INTERESTS

- Technical Writing**: Author of deep-dive articles on ML, Algorithms and Distributed Systems on **Medium** and **Personal Blog**.
- Sports**: Active half-marathon runner (10+ sanctioned events)([Strava](#)) and Tennis
- Creative Pursuits**: (Photography)